Assurance from Junior Allies: Cooperative Management of Adversaries Below the War Threshold

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Abstract

How do allies manage an adversary once all-out war is deterred but provocations continue? Unchecked provocations can escalate to war, so managing them is part of preventing war. I argue that in this setting additional coercion yields little, a condition I call coercion saturation, while cooperative policy that assures the adversary can reduce provocations. The junior ally, not only the patron, can supply such assurance. Using an original monthly dataset of U.S. and South Korean policy toward North Korea from 1991 to 2011, coded for each ally across five dimensions, I find that South Korean cooperative rhetoric and U.S. diplomatic engagement are each independently associated with fewer provocations. The junior ally's effect holds after controlling for the senior ally's full range of instruments. Managing adversaries below the war threshold thus depends not only on the patron's threats but on how each ally assures.

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1 Introduction

States spend the overwhelming majority of their time not at war. Open warfare is the exception in international politics; the norm, even among rivals, is militarized friction short of war, the threats, displays, and limited uses of force that recur without escalating to open conflict (Diehl and Goertz 2000; Palmer et al. 2022). Yet this sub-war friction is not harmless. Major wars rarely erupt from nowhere; they grow out of smaller provocations that are mismanaged and allowed to escalate. The July 1914 crisis is the canonical example: a single assassination, through a chain of reactions, cascaded into a great-power war that few had chosen. If war can emerge from the escalation of lesser conflicts, then managing the sub-war space, keeping provocations from spiraling, is itself central to preventing war. This raises a broader question: once major war is held off, who manages the provocations that remain, and how?

In many of these rivalries, the state facing the adversary is not alone but backed by a stronger ally. Such asymmetric alliances, a powerful patron committed to a weaker partner against a shared adversary, are a common setting for deterred, sub-war conflict. There, who manages provocations becomes a question about two allies, not one actor. Yet alliance and extended deterrence scholarship offers limited guidance here, because it largely treats alliances as unitary actors or centers the senior ally's capabilities and resolve, casting the junior ally as a recipient of protection rather than an active player vis-à-vis the adversary (Waltz 2010; Snyder 2007). This emphasis is well founded on the threat side, where deterrence rests on aggregate capability and the junior ally is outmatched. But threat alone does not make deterrence work. Assurance, its complement, turns on whether the adversary expects restraint to be reciprocated (Schelling 1966; Huth 1988), and this division between threat and assurance maps onto two levels of the relationship. The patron's standing deterrent holds major war off, a near-constant structural threat; below it lies the day-to-day management of provocations, where more flexible policy does the work and where the junior ally is also well placed to act. And because containing those provocations keeps them from

escalating, this sub-war management is not separate from war prevention but part of it.

This paper examines that space. I focus on asymmetric alliances in which escalation to all-out war is already off the table, yet adversaries still engage in recurring hostile actions short of open war. Here the question is not only who keeps major war off the table, but how each ally shapes the adversary's daily decision to provoke, which bears on war prevention because contained provocations are less likely to escalate. Building on the classic insight that effective deterrence pairs firm threats with flexible assurances (Schelling 1966; Huth 1988), I argue that threat and assurance operate at two levels of the relationship. The standing deterrent, the patron's nuclear and conventional weight, holds major war off and changes little over time; the adjustable policy each ally directs at the adversary is what manages provocations below that threshold. Both allies act at both levels, but in the sub-war zone the junior ally, structurally advantaged by proximity, higher issue salience, and frequent interaction with the adversary, can play a meaningful, independent role that patron-centric frameworks would not lead us to expect. This does not mean the junior ally replaces the patron below the threshold; it means the junior ally is no longer sidelined there.

Why should cooperative policy matter at all once a powerful deterrent is already in place? The reason is that in this setting additional coercion faces diminishing returns, a condition I call coercion saturation. By coercion saturation, I mean a condition in which the adversary already expects catastrophic punishment for major escalation, so additional coercive moves change its expectations about ongoing pressure more than its expectations about the costs of war. Once that point is reached, more coercive policies mostly reiterate a threat the adversary already understands, and a posture dominated by coercion can even sour its expectations, signaling that pressure will tighten regardless of restraint (Pauly 2024). Cooperative tools operate on a different margin, communicating that restraint will be acknowledged and potentially rewarded. In other words, under coercion saturation, coercive policies degrade the adversary's expectations of restraint and thus fuel sub-war provocations, whereas cooperative policies improve those expectations and dampen them.

Based on this logic, I advance two claims. First, in asymmetric alliances where all-out war is structurally deterred, more cooperative policy toward the adversary is associated with fewer sub-war provocations, whereas more coercive policy is associated with more provocations. Second, junior-ally policy can independently affect the adversary's behavior.

I test these claims using the U.S.–South Korea alliance and North Korean provocations from 1991 to 2011, a period when all-out war on the peninsula was consistently deterred but sub-war provocations were frequent. I introduce a portable, transparent coding framework and use it to construct a new monthly dataset that codes U.S. and South Korean policy orientations toward North Korea independently across five generic instruments: economic aid, diplomacy, rhetoric, sanctions, and military posture. This granularity captures behavioral variation that aggregate annual measures or binary hawks-versus-doves classifications would miss, and by coding each ally separately it allows a direct test of whether cooperative policies are associated with fewer provocations and whether the junior ally's effect is independent of the senior ally's.

The results are consistent with both claims. At the aggregate level, only South Korea's overall policy orientation shows marginal significance, while U.S. policy and inter-ally alignment are not statistically significant. Disaggregating by dimension reveals that South Korea's cooperative rhetoric and U.S. diplomatic engagement are the only two channels consistently associated with fewer North Korean provocations, and South Korea's rhetorical stance retains its significance even after controlling for the full set of U.S. policy instruments, indicating that the junior ally contributes independently rather than merely transmitting the senior ally's influence. These findings complement the assurance tradition's insight that threats require credible assurances, extending it with a disaggregated picture of how such assurances operate below the war threshold, while challenging the patron-centric emphasis on capabilities and resolve: below the war threshold, cooperative signaling by a junior ally plays a distinct and measurable role in shaping the adversary's decision to provoke. In an exploratory extension using cross-national event data, I apply the same design to other asymmetric alliance triads

and find patterns consistent with the theory, preliminary evidence that the junior ally's role is not unique to the Korean case.

This study advances the empirical study of alliances and deterrence, analyzing how each ally's specific policy instruments independently shape adversary provocations below the war threshold, and building on scholarship that has examined the role of assurances alongside threats, junior-ally agency within alliances, and approaches to measuring alliance contributions (Davis 2000; von Hlatky 2013; Kydd and McManus 2017; Bernhardt and Sukin 2021; Blankenship 2021; Pauly 2024). It makes three further contributions. First, it introduces an original monthly dataset that disaggregates each ally's policy across five instruments; because the coding scheme relies on generic policy categories rather than case-specific knowledge, it is adaptable to other similar settings such as U.S.Taiwan-China. Second, it develops a theory of coercion saturation that explains why cooperative tools reduce provocations while coercive tools face diminishing returns under a fixed strategic deterrent backdrop. Third, it provides evidence that a junior ally can independently shape adversary behavior through its own cooperative policies. These findings also carry a policy implication. Burden-sharing debates often press junior allies to spend more and match the patron's coercive capabilities. Yet a junior ally that helps manage the adversary through its own assurance tools contributes in a different way, and where the patron can count on that contribution below the war threshold, it may free the patron to focus its resources elsewhere.

2 Assurance, Junior Allies, and the Sub-War Gap

Alliance politics builds on balance-of-power and deterrence theory, which conceptualize alliances as devices for aggregating military capabilities to confront external threats and prevent war (Waltz 2010; Walt 1985; Mearsheimer 2001; Brooks and Wohlforth 2005). Conventional approaches therefore emphasize coercive tools such as military capabilities (Huth and Russett 1984; Huth 1988; Leeds 2003; Signorino and Tarar 2006) and resolve, understood

as the willingness to bear costs in defense of allies (Fearon 1995; Fuhrmann and Sechser 2014; McManus 2017), and canonical definitions of deterrence as "the threat or use of force to dissuade an adversary from taking an undesirable action" reinforce this focus on coercion and war-prevention outcomes. Yet classic deterrence theorists have long noted that coercion is not purely about punishment: Schelling highlights that effective coercion combines threats with corresponding assurances that an adversary can avoid pain by complying (Schelling 1966), while Huth's work on extended deterrence shows that a "firm-but-flexible" policy can strengthen deterrence by signaling both determination and restraint (Huth 1988). Subsequent scholarship has developed this insight by treating assurances and promises as integral to influence rather than peripheral to threats (Davis 2000), and by modeling threats and assurances as complementary tools in crisis bargaining (Kydd and McManus 2017). Recent contributions on the assurance dilemma further underscore that deterrence performance depends on how states combine threats with credible assurances, and that mismanaging assurances can backfire even when coercive capability is ample (Pauly 2024). For the most part, though, these insights have been worked out where the stakes are highest—discrete crises with a single defender deciding between war and peace—leaving the everyday management of provocations below that threshold less examined.

This confinement reflects a deeper orientation in the field. Most alliance and extended deterrence research evaluates deterrence in terms of war-versus-peace outcomes and immediate deterrence success. Although rationalist accounts imply a wide bargaining range between full cooperation and all-out war, in which conflict short of war can unfold before bargaining breaks down (Fearon 1995; Reiter 2003), empirical work has overwhelmingly concentrated on whether alliances prevent war rather than on how they manage recurring low-level provocations short of war. This emphasis is closely connected to the field's focus on external balancing and aggregate capability: because alliances are understood as mechanisms for pooling power, it is understandable that scholars have centered the senior ally's military capabilities and deterrent posture. Extended deterrence by definition highlights a powerful patron

that "extends" protection to a weaker protégé (Huth and Russett 1984), and many studies therefore model outcomes as functions of aggregate power mostly provided by senior allies, while junior allies appear mainly as recipients of security worried about abandonment and entrapment (Cha 2000; Snyder 2007; Kudo 2025), as potential free-riders in burden-sharing debates (Sandler and Hartley 1999; Blankenship 2020), or as sources of moral hazard under the alliance umbrella (Benson et al. 2014). A growing literature on small-state agency and asymmetric alliances has challenged the view that junior allies are purely passive, showing that they work to manage autonomy (von Hlatky 2013; Womack 2015), influence patrons (Long 2017), and shape alliance decisions (Wong 2020) despite capability asymmetries. But this work is focused on the intra-alliance relationship rather than on the adversary; it follows naturally from the war-prevention and patron-focused concerns animating the field, yet it leaves open whether and how junior allies independently affect adversary behavior below the war threshold.

Measurement practices reflect the same emphases. Quantitative work on alliances and deterrence typically relies on input measures such as defense spending, spending as a share of GDP, or troop numbers as proxies for contribution to collective security and deterrence (Sandler and Hartley 1999; Blankenship 2021). These aggregate metrics are central to both deterrence and burden-sharing debates, but they are at best noisy proxies for actual deterrent effects (Carroll and Kenkel 2019; Rubin 2023); spending targets like the "2 percent of GDP" benchmark say little about the strategic value of what allies actually do (Kalnins 2017; Alozious 2022). This pattern is understandable: the field has measured deterrence through aggregate inputs largely because it focused on war prevention and patron capabilities, which made input measures seem sufficient. One consequence is that we still lack measures capturing how each ally uses specific policy instruments toward adversaries, making it difficult to see how those tools shape adversary behavior below the war threshold.

The field thus has powerful accounts of how alliances prevent major war, but it says far less about how the junior ally in particular manages sub-war provocations, and prevailing

measures, built on aggregate capability, make that contribution hard to observe. The next section develops a theory of how alliance members, especially the junior ally, shape adversary provocations once war is already deterred.

3 Theory and Hypotheses

3.1 Scope condition: war prevention and sub-war conflict

As discussed above, existing studies on deterrence have made significant progress in analyzing how alliances help prevent war. The primary focus of alliance and extended deterrence research has been on whether and how states avoid all-out war. Yet the world does not operate on a simple war–peace dichotomy; between war and peace lie multiple stages of conflict in which ongoing risk management takes place, and in which states mix coercive threats with assurances in managing adversaries below the war threshold. This sub-war management is not separate from war prevention but an extension of it, since wars typically grow out of provocations left unchecked rather than appearing from nowhere. Most alliance-adversary relationships, moreover, spend the overwhelming majority of their duration in this sub-war zone rather than in open conflict, so it is here that the bulk of adversary interaction actually occurs.

This study focuses on asymmetric alliances in which escalation to all-out war is currently off the table, yet adversaries still pursue limited aggression below the threshold of open conflict. In many contemporary cases, a central structural constraint is the alliance's strategic deterrent posture: the senior ally's nuclear forces and conventional superiority make full-scale war prohibitively costly for the adversary, and the alliance's commitments raise the expected costs of a direct attack on the junior ally to extreme levels. The aim here is not to argue that extended deterrence is always effective or that allies are fully reassured; rather, it is to ask how managing the provocations that recur below the war threshold helps keep major war off the table. This displacement of conflict from the strategic level to the sub-war level reflects

the logic of the stability-instability paradox, in which stability in the strategic balance makes lower-level provocation more likely (Snyder 1965; Jervis 1989).[1] In such settings, adversaries redirect their efforts into sub-war provocations such as missile tests and border incidents, which probe alliance red lines and impose costs without crossing the threshold of open conflict.

Because war typically results from the escalation of tensions rather than emerging out of nowhere, reducing sub-war provocations is itself a contribution to keeping war off the table. This study therefore asks what lowers provocations within the sub-war zone, rather than modeling the point at which provocations would tip into war. Where exactly that tipping point lies is an important question, but it is not the one pursued here. The analysis focuses on periods in which the strategic deterrent keeps major war off the table, while recognizing that this condition is maintained rather than permanent. Within those periods, it examines how alliance members shape the adversary's decision to provoke or refrain by combining coercive threats with assurances about the consequences of restraint.

In addition, this study assumes that the adversary monitors and responds to allied policy choices. While internal dynamics undoubtedly influence adversary behavior, the logic of deterrence presupposes that the adversary is at least partly responsive to external conditions; otherwise, there would be nothing to threaten or assure. Empirical work on North Korean propaganda supports this assumption, showing that Pyongyang's threatening rhetoric responds systematically to changes in its security environment and that surges in such rhetoric precede military provocations (Sukin 2022).

3.2 Coercion saturation and assurances

Assurance has long been recognized as integral to deterrence. Coercion theory holds that a threat works only if it comes with a credible assurance: that complying will spare the

[1] The classic paradox is dyadic and assumes mutual second-strike capability. Here I invoke its central logic, strategic stability displacing conflict downward, rather than its original symmetric setup; recent work extends the paradox to asymmetric and contemporary East Asian settings (Hiim and Tunsjø 2025; Roehrig 2016).

target the punishment (Schelling 1966), and a substantial literature develops this insight (Davis 2000; Kydd and McManus 2017; Pauly 2024). Yet existing accounts treat threat and assurance as two aspects of a single coercive relationship, operating at the same level. This paper stratifies them instead. The underlying strategic structure—the alliance itself and its nuclear and conventional forces, provided mainly by the senior ally—exerts a near-constant threat on the adversary that barely changes as long as the alliance holds and the adversarial relationship persists. What varies is the policy each ally directs at the adversary over time. Because that structural threat is already high and fixed, it supplies the credible backdrop on which assurance depends: adjusting policy in a cooperative direction while the backdrop remains intact reads as assurance, the conditional promise that restraint will be met with restraint, whereas dismantling the backdrop itself would be accommodation or appeasement. By the same logic, adding further coercion onto an already-saturated backdrop yields little, since the adversary expects punishment regardless of whether it shows restraint.

In these environments, the alliance's nuclear and conventional forces already pose an extreme coercive threat, so additional coercive measures face diminishing returns: once a posture credibly threatens catastrophic punishment, each further increment of coercion adds progressively less deterrent value. This has a counterpart in prospect theory's diminishing sensitivity, whereby an adversary already facing catastrophic threatened costs registers little additional pressure from added coercion (Kahneman and Tversky 1979; Tversky and Kahneman 1992). More troops, larger exercises, or additional sanctions mostly reiterate what the adversary already knows: that it is threatened. In this sense, the coercive dimension of the relationship has reached a ceiling, a condition I call *coercion saturation*.

Under coercion saturation, the central uncertainty for the adversary is not whether crossing the war threshold would be costly, but what happens if it shows restraint. Does the alliance keep increasing pressure regardless of the adversary's behavior, or is there a path in which restraint is acknowledged and met with some accommodation? This is fundamentally a problem of the adversary's expectations about how its restraint will be met.

The key distinction is between assurance and mere accommodation. Assurance is conditional: the credible promise that if the adversary shows restraint, punishment will be withheld. It is the complement of a threat, not a substitute for it. Accommodation, by contrast, is an unconditional benefit conferred regardless of the adversary's behavior, the kind of concession the conventional worry treats as weakness inviting further aggression. What separates the two is the deterrent backdrop: when the threat of catastrophic punishment is credible and fixed, a cooperative move is the conditional half of a threat-plus-assurance pair, but absent that backdrop the same move is merely accommodation. Coercion saturation is therefore not only the condition under which added coercion stops working; it is also what converts cooperative policy into credible assurance, since the saturated deterrent ceiling supplies the standing, believed threat that assurance requires.

When coercive moves dominate and cooperative signals are absent, the adversary has little reason to expect that restraint will be rewarded, which can encourage more sub-war provocations rather than fewer. Cooperative policy, by contrast, can function as assurance: it signals that while war-level threats remain in place, the alliance will moderate day-to-day pressure if the adversary refrains, rather than tightening the screws automatically. Threats and assurances are thus complementary, not competing, and they operate at different levels. The war-level threat establishes the high costs of escalation, while variation below the threshold depends on whether the alliance leans coercive or cooperative in its day-to-day policy. Because unchecked provocations can accumulate into escalation, containing them through assurance is more than crisis management; it is part of keeping major war off the table.

> ***H1 (Assurance and cooperative policy).*** *In asymmetric alliances where all-out war is structurally deterred, more cooperative policy toward the adversary, functioning as assurance under the credible deterrent backdrop, is associated with fewer sub-war provocations, whereas more coercive policy is associated with more provocations.*

3.3 The junior ally's independent role

The junior ally faces a security problem its patron largely does not. Even with major war held off by the alliance's deterrent, the adversary keeps probing the junior ally through border incidents, incursions, and other actions that fall short of triggering a full-scale response. Against the adjacent junior ally, such low-level provocation is feasible precisely because it stays below the war threshold; against the distant senior ally, the same kind of act is barely available, since anything able to reach the patron would be large enough to risk war. These provocations are not trivial for the junior ally: they impose military readiness burdens, disrupt the economy, and generate domestic political pressure. The strategic problem facing the junior ally, then, is not whether it will be invaded but how frequently it must absorb provocations just short of war.

Even provocations that target both allies, such as missile and weapons tests, are received differently on each side. For the senior ally, any particular adversary's actions are one regional concern among many global commitments. For the junior ally, they are front-page, high-salience events that demand an immediate response. The alliance's deterrent posture, provided mainly by the senior ally, keeps full-scale war off the table, but it cannot fine-tune the daily risk environment, and the senior ally neither can nor does respond to each provocation with tailored policy maneuvers. This asymmetry of exposure and salience means that the junior ally is the member most directly burdened by sub-war provocations and most motivated to manage them. This does not mean the two allies split the sub-war space between them: the senior ally still acts below the threshold, as the analysis will show. The point is that in this space the junior ally is no longer sidelined, and its own policy choices can independently affect the adversary.

The junior ally is structurally well positioned to manage the adversary below the war threshold. Geographic proximity[2] places the junior ally in continuous contact with the

[2]Examples include Eastern European NATO members facing Russia, the Philippines sharing open sea with China, and South Korea sharing a border with North Korea.

adversary and allows faster reactions than the distant patron; the adversary knows this. Proximity also opens a wider window for frequent interaction across military, diplomatic, and economic channels, generating a steady stream of observable policy moves that the adversary monitors, and it allows the junior ally to accumulate more information about the adversary's intentions and behavior than the distant senior ally can. Beyond this positioning, the junior ally also possesses the capability to act. Though weaker than its patron, it commands its own policy instruments and the independent capacity to act.[3] These means are sufficient to engage an adversary on its own rather than depending on the senior ally to act for it.

Taken together, the junior ally possesses the structural capacity (proximity, frequent contact, informational advantage), the independent means to act, and the strategic motivation (disproportionate exposure to provocations, higher issue salience) to manage adversary behavior below the war threshold on its own. Broadly, the senior ally's posture does most to keep major war off, while the junior ally has the strongest motive to manage the sub-war zone, but the line is not sharp: both act in both zones. The junior ally, in addition to its patron, can shape the adversary's sub-war behavior through its own policy.

> ***H2 (Junior-ally independence).*** *In asymmetric alliances where all-out war is structurally deterred, more cooperative junior-ally policy is associated with fewer sub-war provocations, whereas more coercive junior-ally policy is associated with more provocations, independently of the senior ally's policy.*

4 Data and Methods

In this section, I introduce a novel monthly dataset coding U.S. and South Korean policy orientations toward North Korea across five dimensions, describe how I measure the main outcome of interest, and explain the modeling strategy.

[3] Many formal junior allies are far from negligible powers. Japan, South Korea, and Australia, for instance, rank among the world's larger economies and military spenders (World Bank; SIPRI).

I focus on the US–ROK–DPRK triangle because it is a clear example of an asymmetric alliance, which lets me examine how changes in each ally's policy relate to the adversary's behavior. I structure the dataset as a monthly panel at the month–administration level for the U.S. and South Korea. For each month–administration combination, I code the government's policy orientation toward North Korea across five dimensions. The reasons for my case and time-period choice are threefold. First, I study the period from 1991 onward because it marks the post–Cold War era, when the collapse of the Soviet Union changed the broader strategic environment and prompted adjustments in the US–ROK alliance. This era was marked by recurring missile tests and armed incidents but no escalation to full-scale war, the below-threshold setting the theory addresses. Second, the early 1990s are when North Korea's nuclear and missile programs became central to U.S. and South Korean policy toward Pyongyang, making this a natural starting point for analyzing policy orientations and DPRK reactions. Third, I end the sample in 2011 to isolate the Kim Jong Il era and avoid confounds from the leadership transition to Kim Jong Un, whose tenure reflects different decision-making and a much higher rate of missile and nuclear testing and is better treated as a separate phase.

4.1 Explanatory Variables: A New Five-Dimension Monthly Dataset of Alliance Policy

My main independent variables are monthly policy-orientation measures for each ally. These measures, and the coding scheme that produces them, represent the paper's primary empirical contribution. The dataset advances beyond existing approaches to alliance behavior in two respects.

First, the measures are substantially more granular than existing operationalizations of alliance behavior, which rely on annual or administration-level indicators such as defense spending or troop levels. Monthly coding captures the tempo of below-the-war-threshold interactions between allies and adversaries, which unfolds far faster than annual aggregates can

track. Coding each ally's policy independently, rather than treating the alliance as a unitary actor, makes it possible to test whether the junior ally exerts an independent influence on the adversary: the paper's central question. Furthermore, disaggregating policy into five dimensions allows identification of which specific instruments correlate with adversary responses, avoiding the loss of information that occurs when distinct instruments are collapsed into a single index. The five-point scale within each dimension also captures gradations in policy intensity that a binary hawks-versus-doves framing would miss; for instance, the difference between a moderate conciliatory signal and a landmark peace declaration.

Second, the five dimensions correspond to generic policy instruments that any state employs toward adversaries: economic aid, diplomacy, sanctions, rhetoric, and military posture. Because these categories are not specific to the Korean Peninsula, the dimensional structure is portable to other alliance-adversary triads (e.g., U.S.–Taiwan–China), requiring only the identification of relevant policy chronologies while the categories, coding scale, and decision rules carry over directly. The coding rubrics specify explicit thresholds and decision rules for each dimension, and they rely on observable policy actions rather than subjective assessments of intent, supporting inter-coder reliability testing and straightforward replication.

To construct these monthly measures, I rely primarily on official South Korean and U.S. government sources that provide detailed chronologies of policy actions toward North Korea. For the ROK side, the main sources are the Ministry of National Defense's 국방연표 (Defense Chronology) and the Ministry of Unification's 남북관계연표 (Chronology of Inter-Korean Relations) and 통일백서 (Unification White Paper), which document aid decisions, inter-Korean talks, sanctions measures, official statements, and military posture changes. For the U.S. side, I draw on the same ROK chronologies for externally visible actions, supplemented by English-language chronologies such as those compiled by the Arms Control Association and the National Security Archive at George Washington University, as well as official U.S. government policy and strategy documents for each administration (e.g., National Security Strategies, Defense Department reports, and State Department releases).

Based on these sources, I construct the independent variables by combining two components: a doctrinal baseline that captures each administration's overall intended posture toward North Korea (B_g), and five policy dimensions that capture monthly policy behavior (X_t).

The doctrinal baseline (B_g) represents each administration's stated strategic orientation toward North Korea, coded from the official sources described above. The baseline takes a value of $+1$ when the administration prioritizes engagement and rapprochement, 0 when the doctrine is balanced or ambiguous, and -1 when it emphasizes confrontation and hostility. For example, Kim Dae-jung's "Sunshine Policy" is coded as $+1$, since it was a clearly cooperative era, while Obama's "Strategic Patience" is coded as 0, reflecting a posture that neither prioritized engagement nor confrontation. This baseline is fixed throughout each administration's tenure, capturing doctrinal intent rather than month-to-month behavior. I restrict it to the narrow $[-1, +1]$ range to reflect that it represents an underlying orientation, not an active policy implementation.

However, the doctrinal baseline alone cannot capture the significant variation in policy behavior within a single administration. Even under Kim Dae-jung's strongly engagement-oriented presidency, South Korean policy toward North Korea fluctuated between cooperative and hostile signals depending on events; for instance, in July 1999, Kim urged U.S. Defense Secretary William Cohen to support missiles with ranges over 500 km as a deterrent against North Korea's missile development, a move that also sent a confrontational signal toward Pyongyang. To capture this variation, I code five monthly policy dimensions—economic aid (E), diplomacy (D), sanctions (S), rhetoric (R), and military posture (M)—each ranging from -2 (strongly confrontational) to $+2$ (strongly engagement-oriented). Together, these form the monthly behavioral component:

$$X_t = \frac{1}{5}\Big(E_t + D_t + S_t + R_t + M_t\Big) \tag{1}$$

The overall monthly orientation is then computed as $O_t = B_g + X_t$, combining doctrinal intent with observed behavior in a given month. This formulation allows reinforcement and offsetting effects: an engagement-leaning doctrine combined with confrontational actions may yield a neutral net orientation.

As a general coding rule, policies are coded in the month of implementation, and their effects are assumed to decay in subsequent months unless the action is ongoing or renewed. Each dimension ranges from -2 to $+2$, except for economic aid, which takes only non-negative values since restrictive economic measures are captured under sanctions. Although economic aid and sanctions might appear to be a single axis, I maintain their separation because providing new aid and lifting existing sanctions carry different political and communicative implications, even though both indicate cooperative intent. When multiple events occur within a dimension during the same month, I record the strongest one (highest absolute value). Below, I describe each dimension and illustrate the coding with examples from the dataset.

Economic aid captures officially announced or implemented economic assistance or humanitarian aid, coded on the basis of publicly confirmed commitments or disbursements rather than proposals. Scores range from $+1$ for minor assistance (below \$50 million USD or 50 billion KRW) to $+2$ for major packages (at or above those thresholds).

Diplomacy captures official talks, negotiations, working-level meetings, or summits between the U.S. or South Korea and North Korea, including multilateral formats such as the Six-Party Talks. Higher-level engagement receives higher scores: former President Carter's 1994 visit to Pyongyang scores $+2$, while routine working-level meetings score $+1$. Cancellation or suspension of talks is coded negatively: withdrawal from planned negotiations scores -1, while formal termination of dialogue channels would score -2, though no such case appears in the dataset.

Sanctions captures new unilateral restrictive measures, or official easing and relaxation, by the U.S. or South Korea; UN-initiated sanctions are excluded. This dimension measures

the direction of policy change in a given month, not the standing level of restrictions already in place. Comprehensive sanctions packages score -2, while narrower or targeted measures score -1. Partial relaxation or temporary exemptions score $+1$, and substantial lifting or restoration of trade channels scores $+2$.

Rhetoric captures the tone and content of official government communications, speeches, statements, and designations concerning North Korea. For example, President Roh Tae-woo's July 1991 directive in Vancouver for breakthrough inter-Korean civilian exchanges, issued as part of his Nordpolitik, is coded as $+2$, a landmark conciliatory statement. Conversely, South Korea's designation of North Korea as the "principal enemy" in its December 2010 Defense White Paper is coded as -2.

Military posture captures joint exercises, military deployments, or readiness changes explicitly tied to North Korea, excluding unrelated global operations. Major new deployments or large-scale joint exercises score -2, while moderate expansions of exercise scale score -1. Partial postponement or downsizing of exercises scores $+1$, and complete suspension of joint exercises or troop withdrawals for conciliatory purposes scores $+2$.

Finally, in addition to each ally's individual measures, I construct an alignment variable to test whether policy coordination between the two allies affects adversary behavior independently of each ally's own policy stance. Larger values indicate greater divergence between allies on that dimension. If the conventional view is correct that the alliance deters as a unit, alignment should matter; if instead each ally operates through its own channel, alignment should add little explanatory power. For each dimension, I calculate the absolute difference between the U.S. and South Korean scores. Perfect coordination yields 0; for example, if U.S. rhetoric is -1 and South Korean rhetoric is $+2$ in a given month, the alignment score for that dimension is 3.

4.2 Outcome Variable: North Korean Provocations

The theory developed in Section 3 concerns adversary behavior below the war threshold: hostile actions that impose real costs on the alliance in the form of readiness burdens, economic disruption, escalation risk, and domestic political pressure. I operationalize this concept as the monthly count of North Korean provocations. I define provocation here as a discrete, publicly observable act by the DPRK that signals hostile intent toward the U.S., South Korea, or both, while remaining below the level that would be expected to trigger full-scale military retaliation. This definition encompasses a range of actions, including missile and nuclear tests, exchanges of fire, armed infiltrations, naval incursions, bombings, and similar hostile acts, unified by their shared strategic function: probing alliance redlines and imposing costs without crossing the war threshold.

The data are drawn from the CSIS Beyond Parallel provocation dataset, which records North Korean provocations from 1958 to the present using a consistent event-identification protocol based on official government reports, military announcements, and verified news sources (CSIS Beyond Parallel 2025). The database covers physically implemented hostile acts such as missile and nuclear tests, exchanges of fire, infiltrations, and incursions, and does not code purely rhetorical threats or statements as provocations. I aggregate these individual events into a monthly count for the 1991–2011 sample period.

A count specification rather than a binary provocation/non-provocation indicator is appropriate here because the theoretical question is not simply whether North Korea provokes in a given month but how much adversary activity the alliance faces. If cooperative policy reduces sub-war pressure, that effect should appear as a reduction in the number of provocations per month, a distinction that a binary measure would obscure.

Figure 1 displays the monthly distribution of DPRK provocations over the sample period. Two features of the distribution are immediately apparent. First, the data are dominated by zeros: of 252 monthly observations, 194 (77 percent) register no provocations at all, while the remaining months are concentrated at low counts with occasional spikes. The mean

monthly count is 0.33 and the variance is 0.60, confirming substantial overdispersion relative to a Poisson process. This structure motivates the negative binomial specification used in the estimation. Second, provocations tend to cluster in time, appearing in short bursts separated by long quiet periods rather than arriving at a steady rate. This pattern is consistent with an adversary that responds to shifting external conditions, in line with the assumption that Pyongyang monitors allied policy signals.

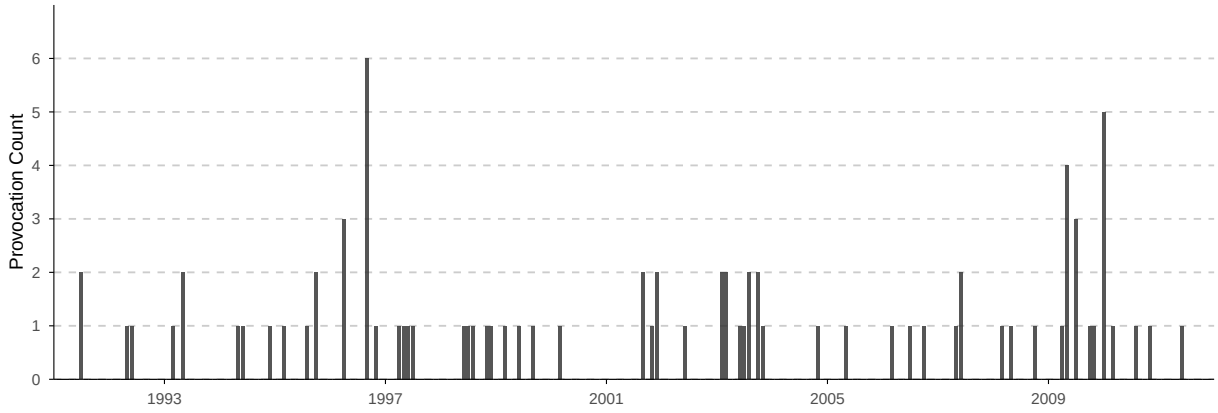


Figure 1: Monthly DPRK Provocation Count, 1991–2011. Data from CSIS Beyond Parallel. Alt text: Bar chart showing monthly counts of North Korean provocations from 1991 to 2011, with most months recording zero and occasional spikes reaching up to six.

4.3 Estimation Strategy

I estimate the relationship between allies' policy orientations and DPRK provocations using negative binomial regression models. As the distribution described above indicates, the dependent variable is overdispersed with a large mass at zero, which violates the equidispersion assumption of the Poisson model and makes the negative binomial specification more appropriate. I use the standard negative binomial rather than a zero-inflated model because the excess zeros in the data have a substantive interpretation: most months produce no provocations precisely because the adversary chose not to act, not because provocations were structurally impossible in those months.

I include all policy-orientation variables with a one-month lag, so that U.S. and ROK

policy choices in month $t-1$ are used to predict DPRK provocation counts in month t. The lag reflects the theoretical assumption that policy signals require time to be observed and processed by the adversary before influencing behavior, and it helps mitigate simultaneity concerns. The robustness checks in Section 5 confirm that effects are concentrated at the one-month lag and disappear at two and three months, consistent with Pyongyang responding to recent signals rather than accumulated policy history. The descriptive statistics above refer to the full 1991–2011 monthly series (252 months); the lagged models drop the first observation, yielding 251 months in the estimation sample.

I estimate separate models using overall policy orientation scores, disaggregated policy dimensions, and alignment measures. First, I estimate bivariate models using each ally's overall policy orientation ($O_t = B_g + X_t$), which combines the administration-level doctrinal baseline with the monthly behavioral component:

$$\log\Big(\mathbb{E}[\text{Provocations}_t]\Big) = \beta_0 + \beta_1 \, O_{t-1}^{\text{ROK}} \tag{2}$$

$$\log\Big(\mathbb{E}[\text{Provocations}_t]\Big) = \beta_0 + \beta_1 \, O_{t-1}^{\text{US}} \tag{3}$$

$$\log\Big(\mathbb{E}[\text{Provocations}_t]\Big) = \beta_0 + \beta_1 \, \text{Alignment}_{t-1} \tag{4}$$

The disaggregated model enters all fifteen dimension-level regressors simultaneously, where each dimension enters as a separate regressor with its own coefficient:

$$
\begin{aligned}
\log\Big(\mathbb{E}[\text{Provocations}_t]\Big) = \beta_0 \\
&+ \beta_1 \,\text{ROK}_{E,t-1} + \beta_2 \,\text{ROK}_{D,t-1} + \beta_3 \,\text{ROK}_{S,t-1} \\
&+ \beta_4 \,\text{ROK}_{R,t-1} + \beta_5 \,\text{ROK}_{M,t-1} \\
&+ \beta_6 \,\text{US}_{E,t-1} + \beta_7 \,\text{US}_{D,t-1} + \beta_8 \,\text{US}_{S,t-1} \\
&+ \beta_9 \,\text{US}_{R,t-1} + \beta_{10} \,\text{US}_{M,t-1} \\
&+ \beta_{11} \,\text{Align}_{E,t-1} + \beta_{12} \,\text{Align}_{D,t-1} + \beta_{13} \,\text{Align}_{S,t-1} \\
&+ \beta_{14} \,\text{Align}_{R,t-1} + \beta_{15} \,\text{Align}_{M,t-1}
\end{aligned}
\tag{5}
$$

In additional robustness checks reported in the Results section, I examine alternative lag structures and specifications, including tests of co-movement, linearity, and reverse causality tests. The disaggregated model enters only the fifteen dimension-level regressors without additional controls; I discuss this design choice and potential confounds in Section 5.4.

Figure 2 shows the monthly variation in US and ROK policy across the aggregate orientation and all five dimensions over the full 1991–2011 sample period. The plots demonstrate considerable fluctuation within each ally over time and illustrate the divergence between the two allies, suggesting that the junior ally maintained an independent policy orientation distinct from its patron's. It is also noticeable that sanctions were very rarely used: South Korea recorded only two sanctions-related policy actions toward North Korea over the entire twenty-year period (Ministry of Foreign Affairs 2026).

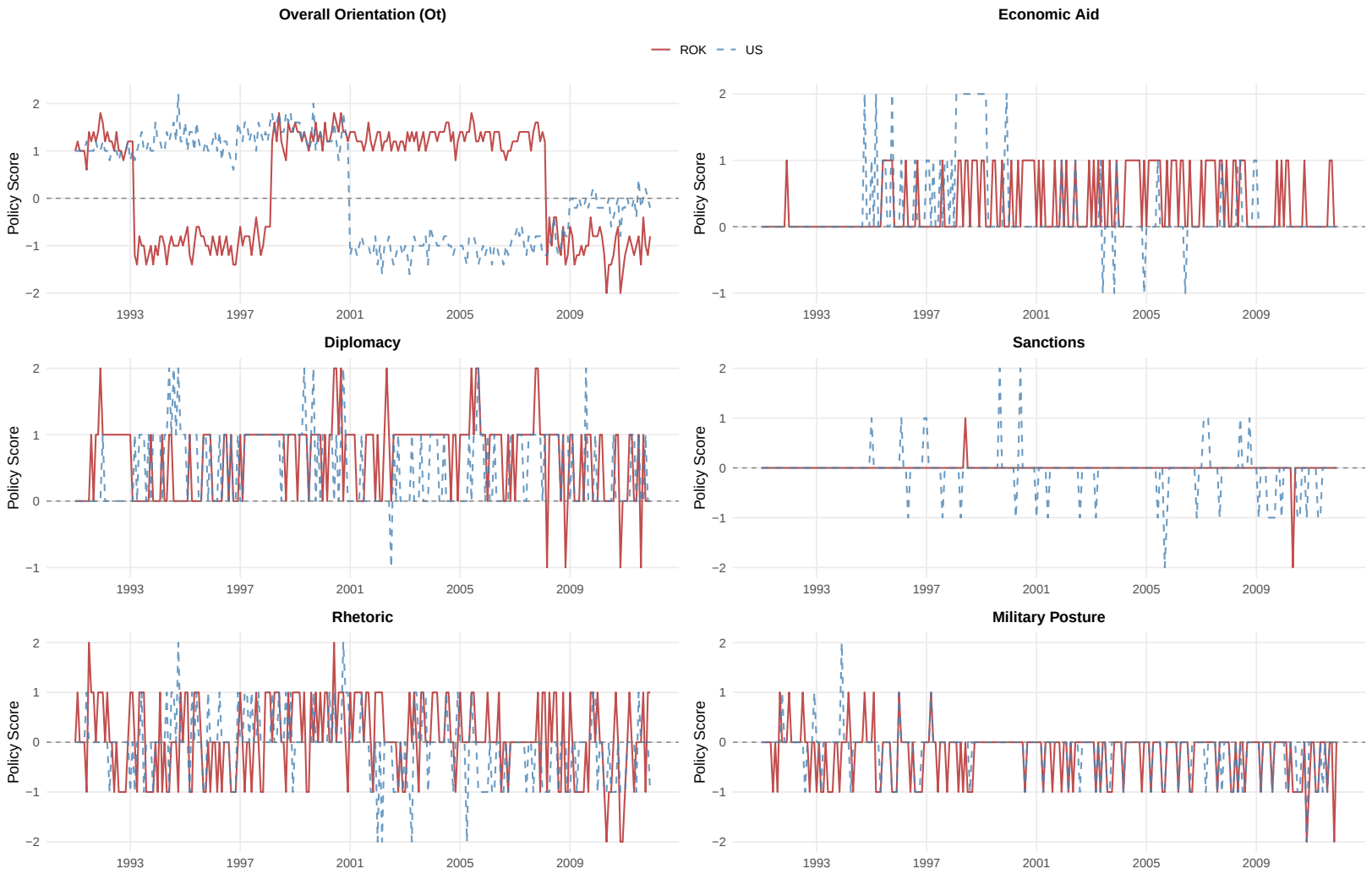


Figure 2: US and ROK Policy Toward North Korea by Dimension, 1991–2011. Positive values indicate engagement-oriented policy; negative values indicate confrontation-oriented policy. Alt text: Six line plots showing monthly US and ROK policy scores toward North Korea from 1991 to 2011 across overall orientation, economic aid, diplomacy, sanctions, rhetoric, and military posture. ROK scores are shown in solid red and US scores in dashed blue.

5 Results and Discussion

5.1 Cooperative policy

H1 expects that more cooperative policy is associated with fewer provocations, whereas more coercive policy is associated with more. The aggregate orientation models offer a first test. When each ally's overall policy orientation (O_t) is entered as a single predictor, the ROK coefficient approaches conventional significance ($p = 0.092$) in the negative direction, indicating that a more cooperative overall posture is associated with fewer provocations (Table 1).

Table 1: Overall Orientation and DPRK Provocations

	ROK	US	Alignment
Overall Orientation (O_t)	$-0.200^{\dagger}$	-0.022	0.123
	(0.119)	(0.130)	(0.160)
Constant	-1.056^{***}	-1.091^{***}	-1.311^{***}
	(0.140)	(0.141)	(0.318)
N	251	251	251

† p <0.1, * p <0.05, ** p <0.01, *** p <0.001
Entries are log-coefficients from negative binomial models. Standard errors in parentheses. Exponentiate coefficients to obtain incidence rate ratios. All independent variables are lagged one month. Dependent variable: monthly DPRK provocation count. Sample: 1991-2011. $O_t = B_g + \bar{X}_t$.

However, because O_t collapses five distinct instruments into a single score, it cannot isolate which specific policy channels are driving this pattern. Disaggregating by dimension clarifies the picture. In the multivariate model with all fifteen dimension-level regressors entered simultaneously (Table 2), a one-unit increase in South Korea's cooperative rhetoric is associated with approximately a 31% reduction in expected DPRK provocations, while a one-unit increase in U.S. diplomatic engagement is associated with approximately a 55% reduction (Figure 3).

Table 2: All Policy Dimensions and DPRK Provocations

		Estimate
ROK	Economic Aid	0.043 (0.379)
	Diplomacy	−0.068 (0.259)
	Sanctions	1.660 (1.544)
	Rhetoric	−0.364* (0.182)
	Military Posture	−0.143 (0.385)
US	Economic Aid	0.081 (0.268)
	Diplomacy	−0.797** (0.287)
	Sanctions	−0.651 (0.549)
	Rhetoric	0.320 (0.222)
	Military Posture	0.071 (0.437)
Align	Economic Aid	0.269 (0.336)
	Diplomacy	−0.135 (0.262)
	Sanctions	−0.267 (0.558)
	Rhetoric	0.111 (0.215)
	Military Posture	−0.199 (0.415)
	Constant	−0.898** (0.326)
	N	251

† p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Entries are log-coefficients (standard errors in parentheses) from negative binomial models. Exponentiate coefficients to obtain incidence rate ratios. All independent variables are lagged one month. Dependent variable: monthly DPRK provocation count. Sample: 1991–2011.

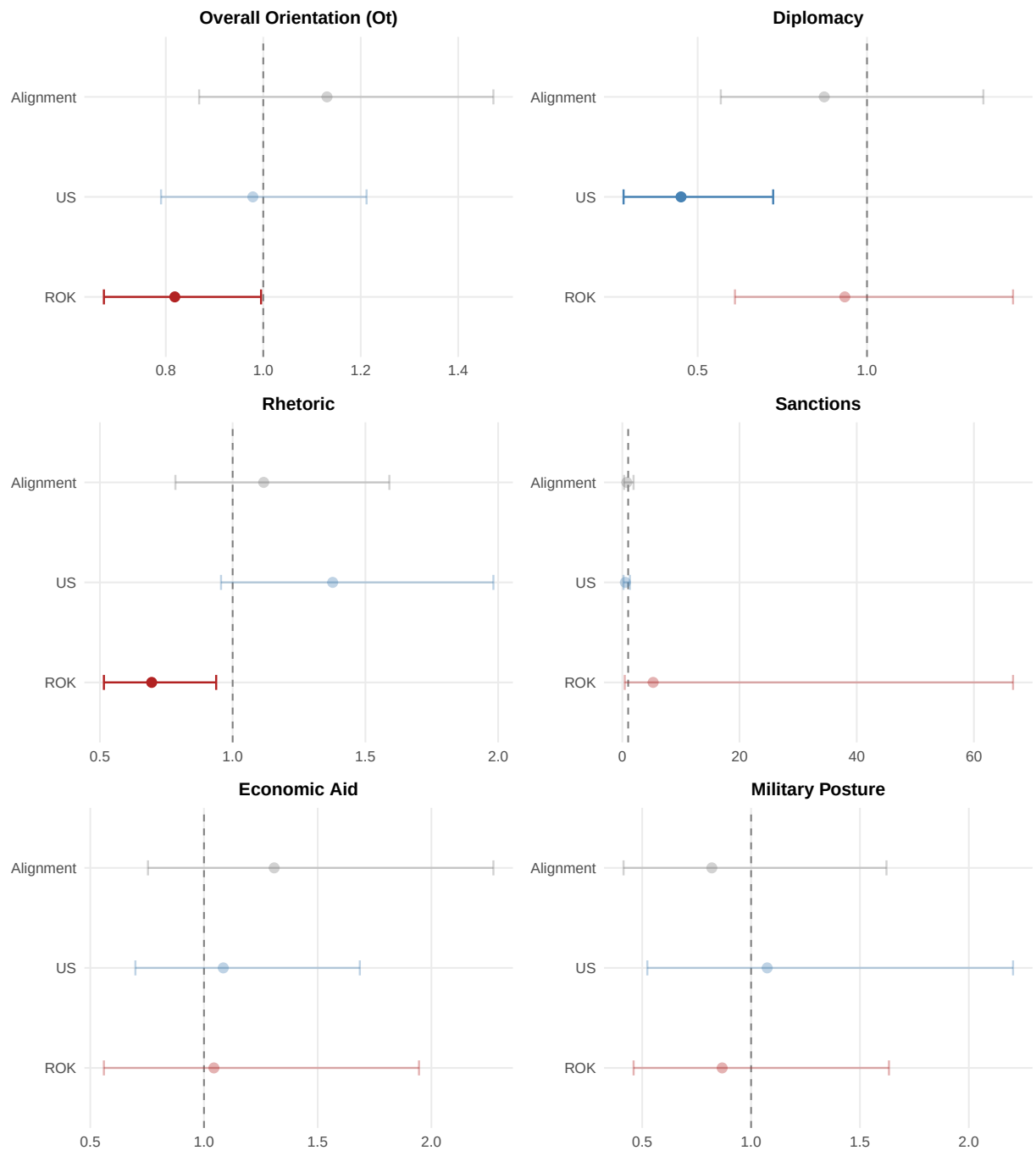


Figure 3: Policy Dimensions and DPRK Provocations. Incidence rate ratios with 90% confidence intervals (IRR < 1 = fewer provocations; IRR > 1 = more provocations). Alt text: Dot plot showing incidence rate ratios with 90% confidence intervals for ROK, US, and alignment across six policy dimensions. Values below one indicate fewer provocations. ROK rhetoric and US diplomacy show statistically significant negative effects.

The remaining dimensions, economic aid, sanctions, and military posture, show no sig-

nificant effects for either ally. The two significant channels survived multiple robustness checks, including tests for co-movement, reverse causality, and alternative lag structures. Both were lagged one month and were not predicted by prior provocations in the reverse causality tests, strongly suggesting that cooperative South Korean rhetoric and US diplomacy preceded, rather than responded to, lower provocation rates from North Korea. This pattern supports H1: under a credible strategic deterrent backdrop, cooperative policy on specific channels that can open an exit for the adversary is associated with fewer provocations, whereas the aggregate orientation measure is not. It is consistent with the assurance literature's emphasis that deterrence depends not only on threats of punishment but on assurances that restraint will be acknowledged (Stein 1985; Kydd and McManus 2017; Pauly 2024). The diplomacy result also converges with CSIS Beyond Parallel's finding of an inverse correlation between US–DPRK negotiations and North Korean provocations (Collins 2017). Why South Korean rhetoric and US diplomacy in particular, rather than the other instruments, is a question I take up in Section 5.3.

The null findings on military posture and sanctions warrant a cautious reading. The models estimate each instrument's average association with provocations, so a null coefficient indicates only that these dimensions show no measurable association with provocations in this sample, in either direction. This is not evidence that these instruments are inert, and the caution cuts both ways: the significant results establish an association along the cooperative-coercive continuum for two channels, not a general effect of coercion or cooperation across all five. This caution is especially warranted for sanctions, which South Korea deployed only twice over the twenty-year period, leaving the estimate imprecise. With that qualification, the insignificance of military posture is consistent with CSIS Beyond Parallel's independent finding that military exercises have no discernible effect on DPRK provocation patterns (Cha et al. 2016), as well as with more recent work showing that joint military exercises fail to deter and may even provoke North Korean hostile responses (Bernhardt and Sukin 2021). The CSIS study also found that the state of US–DPRK diplomatic relations before

exercises was a better predictor of North Korean behavior than the exercises themselves, further reinforcing the primacy of diplomatic context over military signals.

5.2 Junior ally independence

H2 expects that more cooperative junior-ally policy is associated with fewer provocations, independently of the senior ally's policy. The aggregate results already point in this direction: of the three overall orientation models, only South Korea's coefficient approaches significance, while the U.S. and alignment coefficients are substantively small and statistically null (Table 1). The disaggregated model sharpens this finding. South Korea's cooperative rhetoric remains statistically significant in a specification that simultaneously includes all ROK, US, and alignment dimensions (Table 2). That is, even after controlling for the full range of US policy instruments—economic aid, diplomacy, sanctions, rhetoric, and military posture—South Korea's rhetoric toward North Korea retains its independent association with lower provocation rates. This finding challenges the conventional wisdom that junior allies are passive recipients of security rather than active contributors to adversary management.

The mechanisms through which the junior ally can independently influence the adversary's strategic behavior are geographic proximity and the higher salience of adversary management for the junior ally compared to its senior partner. South Korea shares a land border with North Korea while the US is geographically distant, and managing Pyongyang is a top-tier security priority for Seoul in a way it is not for Washington, which faces competing global commitments. Pyongyang understands this asymmetry: North Korea recognizes that South Korea is directly affected by its strategic moves and that Seoul actively pursues its own policy toward Pyongyang rather than simply acting as a US proxy.

The 2001–2008 period illustrates this dynamic concretely. When the Bush administration adopted a confrontation-leaning posture toward Pyongyang, marked by the January 2002 "Axis of Evil" designation alongside sustained hostile rhetoric, expanded sanctions, and the suspension of the Agreed Framework, South Korea's Kim Dae-jung and Roh Moo-

hyun administrations moved in the opposite direction. Seoul continued to frame North Korea as a partner for engagement, emphasizing dialogue and inter-Korean cooperation, even as Washington labeled the DPRK a threat. This produced a sustained period of policy divergence visible in Figure 2. The clash came in June 2002, when a North Korean patrol boat opened fire near the Northern Limit Line in an exchange that killed six South Korean sailors and sank a patrol boat, a deadly incident that could have spiraled into wider conflict (Michishita 2010). Yet it did not. Seoul held its engagement posture rather than turning to confrontation, and within weeks Pyongyang sent a message of regret and called for joint efforts to prevent recurrence, its first public expression of apology since 1976. Provocations did not follow either: the six months after June 2002 recorded none at all. If Pyongyang responded only to the senior ally's posture, US hostility over these years should have driven provocations upward; that they did not is hard to square with a patron-centric account. This is not to reduce North Korea's behavior to allied policy alone. Allied policy still tracks the pattern closely, even alongside other influences.

The null results from alliance alignment reinforce this point from a different angle (Table 1 and Table 2). The fact that alignment is null while ROK rhetoric alone is significant suggests that it is the junior ally's own policy stance, not its coordination with the senior ally, that shapes adversary behavior. Not only does ROK policy survive controlling for US policy, but the degree of policy divergence between the two allies adds nothing to the model.

5.3 Division of labor

The preceding results suggest that cooperative policy is associated with fewer provocations and that the junior ally contributes independently, but they raise a further question: why rhetoric from South Korea and diplomacy from the US, not any other tools? I interpret this pattern as a division of labor between the senior and junior allies. Diplomacy is a higher-profile, resource-intensive tool (summits, envoys, formal negotiations) while rhetoric is a more granular, everyday signal; and this maps onto the structural difference between

what the senior ally and junior ally each bring to the table. The pattern is consistent with a division of labor in which U.S. diplomacy helps set the strategic framework, while South Korean rhetoric helps manage the day-to-day political temperature.

Specifically, US diplomacy matters because it is the one US instrument that directly addresses what North Korea cares about most from Washington: security guarantees and regime recognition (Byman and Lind 2010; Howell 2020). Formal talks and summits signal that the US is willing to engage on Pyongyang's existential concerns. North Korea's primary grievance with the US is strategic and security-oriented, so diplomatic engagement is the channel through which Washington can credibly signal that restraint will be rewarded. ROK rhetoric, on the other hand, matters because South Korea is the geographically proximate, directly affected party whose day-to-day statements Pyongyang monitors as an indicator of the immediate security environment. ROK rhetoric is low-cost, high-frequency, and adjustable month-to-month, making it well suited to managing sub-war dynamics. Rhetoric is often characterized as "cheap talk" because it is costless to produce and non-binding. Yet as Sartori (2002) argues, diplomatic communication carries reputational costs that give it credibility beyond its material price. The finding that ROK rhetoric is associated with fewer provocations is consistent with this logic: when the adversary is already monitoring the proximate ally for any indication of the immediate security environment, the content of those signals matters because misrepresentation would damage Seoul's credibility in future interactions with Pyongyang. Pyongyang treats Seoul's rhetorical posture as an indicator of whether the immediate neighborhood is hospitable or hostile to its interests.

Furthermore, the Granger causality and cross-correlation tests showed that these two channels move independently rather than in coordination, which is consistent with the alignment null result. Each ally operates through its own channel, not through joint action. Both ROK rhetoric and US diplomacy are among the most frequently deployed instruments in their respective toolkits, which means they also generate the most consistent stream of signals for Pyongyang to observe.

5.4 Robustness and additional specifications

To address concerns that temporal dependence in provocations may be driving the results, I estimate models that incorporate lagged outcomes and alternative lag structures. Two findings emerge. First, adding the prior month's provocation count as a control does not change the significance of ROK rhetoric or US diplomacy, and the lagged dependent variable itself is insignificant, suggesting that serial correlation is not driving the results. Second, effects are concentrated at the one-month lag and disappear at two and three months. This pattern is consistent with Pyongyang responding to recent signals rather than accumulated policy history, reinforcing the signaling interpretation. The core findings are also robust to tests for co-movement between the two significant channels, linearity, and multicollinearity; full results appear in the Appendix.

The disaggregated model is deliberately parsimonious, entering only the fifteen dimension-level regressors without additional controls. Many variables that might appear to be omitted—such as diplomatic engagement, sanctions regimes, or military exercises—are already captured within the five-dimension coding scheme, which was designed to be comprehensive. The doctrinal baseline (B_g) absorbs administration-level variation on the allied side through the aggregate orientation models, and the sample restriction to 1991–2011 effectively holds constant the DPRK leadership era by excluding the structurally different Kim Jong-un period.

To address potential confounds external to the coding scheme, I test a binary indicator for months containing major DPRK national holidays and a linear time trend; both are insignificant in bivariate specifications, and adding them to the full model individually or in combination does not change the significance of ROK rhetoric or US diplomacy (Appendix). China's influence is partially captured through its downstream effects: when Beijing facilitates or obstructs multilateral diplomacy, e.g., the Six-Party Talks, the observable consequences register in the US and ROK diplomacy dimensions. Taken together, the comprehensive coding scheme, the doctrinal baseline, the sample restriction, and the hol-

iday and time-trend tests address the most important confounds. While these checks do not eliminate the possibility of omitted variable bias, they suggest that the results are not straightforwardly attributable to an unobserved third factor.

5.5 Beyond the Korean case

To probe whether this pattern extends beyond the Korean case, I turn to event data from the Integrated Crisis Early Warning System (ICEWS) (Boschee et al. 2015), which differs from my hand-coded measure in both source and method: it is machine-coded from news reports and captures the overall conflict–cooperation tone of each side's behavior. Applying an analogous monthly design to 32 asymmetric alliance triads from 1995 through 2011, each pairing a U.S. treaty ally or close security partner with a shared adversary, I find that in nearly every case where the association is statistically significant, more cooperative behavior by one side predicts a more cooperative response from the adversary the following month. The junior ally's orientation is independently and significantly associated with the adversary's response in 13 of the 32 cases, controlling for the senior ally, including clear frontline pairings such as South Korea and Japan facing North Korea and Poland and Lithuania facing Russia. The Korean replication closely tracks the hand-coded result, confirming that the finding is not an artifact of my coding. This cross-case evidence suggests that the junior ally's independent influence observed in the U.S.–ROK case is not idiosyncratic. Full results for all 32 triads and the scope conditions governing case selection appear in the online appendix.

6 Conclusion

This paper has asked how allies' policy orientations toward the adversary shape provocations in asymmetric alliances once all-out war is already off the table, and whether junior allies exert an influence of their own. Focusing on the U.S.–ROK alliance and North Korean provocations from 1991 to 2011, it argued that in this below-the-war-threshold space,

assurance-bearing cooperative policy is associated with fewer provocations while added coercion is less likely to suppress them, and that the junior ally's policy can shape this likelihood independently of the senior ally. Using an original monthly dataset that disaggregates U.S. and South Korean policies across five instruments, the analysis showed that at the aggregate level, only South Korea's overall orientation approaches significance, while U.S. policy and inter-ally alignment show no effects. Disaggregating by dimension, South Korea's cooperative rhetoric and U.S. diplomatic engagement are the only channels consistently associated with fewer North Korean provocations. South Korea's rhetorical stance remains significant even after controlling for the full range of U.S. policy instruments, indicating that a junior ally can independently influence adversary behavior through its own cooperative policies rather than merely transmitting the patron's posture.

These findings speak directly to the gaps identified in Section 2 and carry two main implications for scholarship on alliances and extended deterrence. First, they show that assurance, not only threat, is central to managing an adversary below the war threshold: once coercive pressure is saturated, what shapes provocations is cooperative policy that makes restraint worthwhile for the adversary. Second, they show that assurance need not come from the patron. Because the junior ally is structurally suited to provide it, disaggregating both allies and their policy instruments reveals channels of assurance that treating alliances as unitary actors, or focusing on the senior ally's capabilities and resolve, would obscure. In this sub-war setting, deterrence depends not only on the patron's ability to threaten punishment but on how each ally assures the adversary that cooperation will be returned.

The analysis also points to concrete policy implications for alliance management and burden sharing. Current burden-sharing debates typically center on aggregate inputs such as defense spending levels or troop contributions, implicitly treating more capability as the primary way junior allies contribute to deterrence. The evidence here suggests a different division of labor may be more efficient: senior allies can continue to provide the bulk of coercive capability and high-profile diplomacy, while junior allies play a deliberate role in

day-to-day cooperative engagement, especially through rhetorical signaling. Rather than pushing junior allies only to spend more or match the patron's coercive posture, policymakers could instead encourage and empower them to use engagement instruments that they are structurally well positioned to wield, thereby improving deterrence performance below the war threshold without simply scaling up aggregate expenditure.

A few limitations remain. All provocations are weighted equally, though future work could differentiate between types of actions, for example assigning greater weight to nuclear tests than to minor incursions. In addition, the sample ends in 2011 and therefore does not capture the Kim Jong Un era, whose different decision-making and strategic context warrant separate treatment; this structural break is the reason for the 2011 cutoff. These constraints mean the paper cannot claim that junior-ally cooperative policy will always reduce provocations across alliances and time, but they do show that under the conditions studied, junior allies can play a meaningful and previously overlooked role.

The ICEWS probe indicates the pattern is not unique to Korea; the natural next step is to test it with the same disaggregated monthly coding in other cases. A first candidate is the U.S.–Taiwan relationship. Although Taiwan lacks a formal alliance with the U.S. and its statehood remains contested, the structural conditions that drive the theory are present: geographic proximity to the adversary, high issue salience, frequent cross-strait interaction, and a de facto U.S. security commitment through the Taiwan Relations Act. Taiwan's independent policy swings between engagement and confrontation toward Beijing, China's sustained sub-war provocations, and a high-cost escalation environment closely mirror the conditions on the Korean Peninsula; applying disaggregated monthly coding to Taiwanese and U.S. policies toward China would allow a direct test of whether the framework travels beyond formal alliances to de facto security partnerships (Bush 2005; Stokes 2023).

Across these and other extensions, the core message remains the same: to understand how adversaries are managed below the war threshold, we must look not only at what powerful patrons do, but also at how junior allies use their own cooperative policies to shape adversary

behavior day to day. And because unchecked provocations are how major war most often begins, managing them is not separate from preventing war but part of it. Recognizing the junior ally's role here helps move debates about alliance politics and extended deterrence beyond aggregate capabilities, toward a more precise understanding of how specific allies, through specific policies, shape adversary behavior below the war threshold and, in doing so, help keep major war off the table.

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